

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **6520**

L 61, 47A, 986, 1, 1505, 48289

AV:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	202	Seat No:	2022600021
		Course Section:	1		



Name of the candidate. Madhura Kanfode

Candidate's UID number:

2	0	2	2	6	0	0	0	2	1
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Examination class room number:

2	0	2
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Invigilator's signature with date: [Signature] 08/10/2025

(To be filled in by the candidate)

EXAMINATION: BE BTECH MSE
(For example FY MCA/FY BTECH)

SEMESTER : VII

PROGRAM: AIML

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: 8/10/25 Evening

COURSE NAME: Deep Learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Q.1a

$$f(w) = \frac{1}{2} \|Xw - y\|^2$$

$$f'(w) = X^T X w - X^T y$$

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4

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Q.1b

- 1] Expected squared error tell us difference between training and testing errors
- 2] Loss functions like MSE are used to calculate and improve model accuracy using actual and predicted output.
- 3] If training error is less and testing error is also less then it means model is accurate
- 4] If training error is less and testing error is high then expected squared error increases stating model is overfitted and high variance
- 5] If training error is high and testing error is high then expected squared error increases stating model is underfitted and high bias.
- 6] overfitting happens when model capacity too high because it learns every parameter and noise also
- 7] Because of the noise learned when we test data we get wrong outputs.

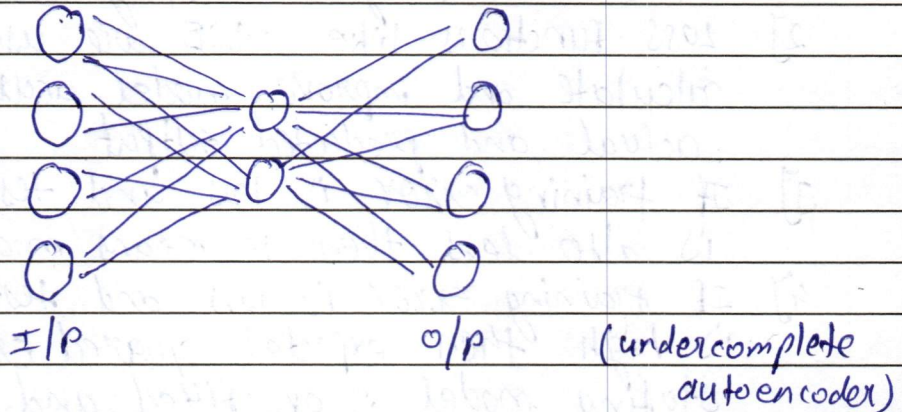
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Q26

Q26



- 1] PCA only works for linear data
- 2] If we train linear undercomplete autoencoder with MSE then ~~the~~ encoder will convert data into less ~~dimensional~~ dimensional data.
- 3] decoder will learn to convert that data into original outputs.
- 4] In this case reconstruction error is calculated using MSE
- 5] In PCA also we convert high dimensional data into low dimensional data trying to keep as much parameter as we can.
- 6] Both model are basically performing dimensionality reduction on linear data using difference between predicted and actual values to keep parameter in check
- 7] overcomplete autoencoder has more number or equal no. of neurons in bottleneck compare to input
- 8] So if we don't perform regularization input data will have more neurons in bottleneck so there will be no dimensionality reduction
- 9] Instead autoencoder will simply copy input into output without learning anything
- 10] This will lead to learning identity function

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Q.3a

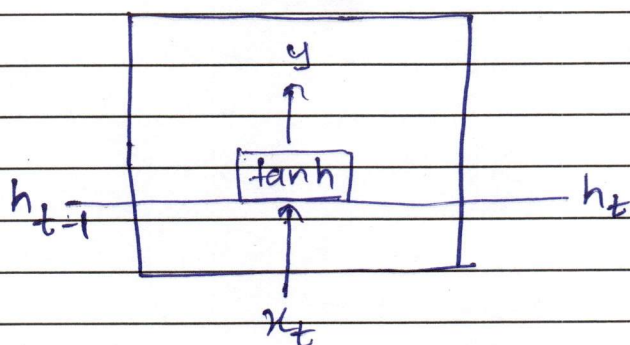
$$\frac{\partial L}{\partial h_{t-k}}$$

1] In RNN $w_{new} = w_{old} - \eta \frac{\partial L}{\partial w}$

$$L = \frac{1}{n} \sum (h_t - \tilde{h}_t)^2$$

$$h_t = \tanh(w_h[h_{t-1}, x_t] + b_h)$$

2] $\frac{\partial L}{\partial w} = \frac{\partial L}{\partial h_t} \times \frac{\partial h_t}{\partial w}$



$$y = \tanh(w_y h_t + b_y)$$

3] $\frac{\partial L}{\partial h_{t-k}} = \frac{\partial L}{\partial h_t} \times \frac{\partial h_t}{\partial h_{t-1}} \times \dots \times \frac{\partial h_{t-k+1}}{\partial h_{t-k}} \quad \text{--- (1)}$

$$\frac{\partial L}{\partial h_{t-k}} = \frac{\partial L}{\partial w_y} \times \frac{\partial w_y}{\partial w_h} \times \frac{\partial w_h}{\partial h_t} \times \frac{\partial h_t}{\partial h_{t-1}} \times \frac{\partial h_{t-1}}{\partial h_{t-k}}$$

4] In LSTM, $c_t = f_t c_{t-1} + i_t \tilde{c}_{t-1}$
i.e. forget gate value will tell if information from last memory cell should be maintained or not.

5] In case of RNN as you can see in eq 1 with every sequence the context of previous state decreases. i.e. their value becomes less

6] In case of LSTM we have different memory cell which stores context of previous states which are controlled by forget gate
 $f_t = \tanh[w_f(h_{t-1}, x_t) + b_f]$



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$$\frac{\partial L}{\partial w} = \sum \frac{\partial L}{\partial h_t} \times \frac{\partial h_t}{\partial h_{t-i}} \times \frac{\partial h_{t-i}}{\partial w}$$

value of i increases



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3b.

1] hidden state in GRU

$$h_t = (1 - z_t) h_{t-1} + z_t \tilde{h}_t$$

$$\tilde{h}_t = \tanh [r_t * w_h [h_{t-1}, x_t] + b_h]$$

2] update gate : $z_t = \sigma [w_z [h_{t-1}, x_t] + b_z]$ 3] reset gate : $r_t = \sigma [w_r [h_{t-1}, x_t] + b_r]$ 4] as per the eqn of h_t you can see
if $z = 1$ then $h_t = \tilde{h}_t$ if $z = 0$ then $h_t = h_{t-1}$ as z_t is sigmoid its value will
vary from ~~(0, 1)~~ (0, 1)so its convex combination of h_{t-1} & $\tilde{h}_t$

5.] $z_t = \sigma [w_z [h_{t-1}, x_t] + b_z]$

$$r_t = \sigma [w_r [h_{t-1}, x_t] + b_r]$$

 $\tilde{h}_t =$ candidate hidden state

6] for example lets take

input values $\Rightarrow 4$ hidden values $\Rightarrow 2$

now LSTM has 3 gates and main output

so No. of parameter = $4 (i \times h + h \times h + h)$

$$= 4 (4 \times 2 + 2 \times 2 + 2)$$

$$= 4 (8 + 4 + 2)$$

$$= 4 (14)$$

$$= 56$$

for GRU $\Rightarrow 2$ gates and main outputNo. of parameters = $3 (i \times h + h \times h + h)$

$$= 3 (14)$$

$$= 42$$

No. of parameters $\Rightarrow$ LSTM > GRU7] Because LSTM has 3 gates and extra
memory storage its has more parameters.

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